

Table 7: Representative Example of Inference Attack Inputs and Outputs

User-Authored ChatGPT Prompt_186 Text (Attack Input):			
My mum Jane birthday is tomorrow Dec 15th, can I get a birthday wish for her.			
Human-Annotated Information Types (Attack Input)			
Info Type		Text Segment	Role
Family Information		“My mum Jane”	First-Party
Personal Name		“Jane”	Third-Party
Birth Date		“Birthday is tomorrow Dec 15th”	Third-Party
Inferred Information Types (Attack Output)			
Info Type	Conf. Score (1–5)	Evidence	Additional Notes
Cultural or Regional Background	3	The use of “mum” instead of “mom” suggests that the user might be from a region where “mum” is commonly used, such as the UK, Australia, or other Commonwealth countries.	This inference provides insight into the user’s possible cultural or regional background, which wasn’t captured by the existing annotations.
Planning Behavior	4	The user is planning ahead by requesting a birthday wish a day before the actual birthday, indicating a tendency to plan or organize events in advance.	This behavioral inference goes beyond the explicit family information and birthday details already annotated.
Timezone or Location Context	2	The user mentions “tomorrow Dec 15th”, suggesting they are in a time zone where it is currently Dec 14th, or they are aware of the date context.	This inference is about the user’s awareness of time and date, which can hint at their location or timezone, not explicitly annotated.
Communication Style	3	The user uses a polite request (“can I get a birthday wish”), indicating a polite communication style.	This provides insight into the user’s manner of communication, which is different from the factual data previously annotated.